DATA ANALYTICS FOR BUSINESS

with Lizhen Xu, Ph.D.

Text Mining: Overview

Learning Objectives

- Describe the bag of words approach
- Familiarized with the usual steps of preprocessing text
- Calculate TF-IDF
- Represent documents in concepts/topics

Text Mining

- Data Representation
  - Text: an example of *unstructured* data
  - Text data are usually *dirty*
    - Synonyms, homographs
    - Terminology, abbreviations
    - Misspelling, ungrammatical writing, arbitrary abbreviation, etc.
  - Text is heavily *context* dependent
    - How to *understand* the meaning of the text
    - Linguistic structure
    - Word order
    - Ambiguity resolution

Text Mining (Cont'd)

- Text Mining Tasks
  - Classification: e.g., spam filtering
  - Prediction: e.g., expected repair cost/time for a service request
  - Clustering: e.g., cluster medical documents
  - Information Retrieval (IR): e.g., search engine
  - Natural Language Processing (NLP): e.g., machine translation
  - Large Language Models (LLM): e.g., ChatGPT, generative AI

Data Representation

- Terminology
  - ***Tokens*** (or ***terms***)
    - *Processed* words
    - The basic unit of analysis
  - ***Documents***
    - One piece of text, no matter how large or small
    - E.g., a single sentence, a 100-page report, a customer comment
    - All text of a text is retrieved as a *single item* when matched or categorized
  - ***Corpus***
    - A collection of documents

Bag of Words

- **Bag of Words**
  - Ignore grammar, word order, sentence structure, and punctuation
  - *Bag* vs. *Set*
    - Set: allows only one instance of each item
    - Bag: multiset, members are allowed to appear more than once
    - Both ignore the order
  - **Term-Document Matrix**
    - Each row is a term; each column is a document

```
S1.  this is the first     sentence!!
S2.  this is a second Sentence :)
S3.  the third sentence, is here
S4.  forth of all sentences
```

	Docs			
Terms	1	2	3	4
all	0	0	0	1
first	1	0	0	0
forth	0	0	0	1
here	0	0	1	0
second	0	1	0	0
sentence	0	1	0	0
sentence!!	1	0	0	0
sentence,	0	0	1	0
sentences	0	0	0	1
the	1	0	1	0
third	0	0	1	0
this	1	1	0	0

Preprocessing Text

- Tokenization
  - Use delimiters to divide a document into tokens
  - Remove white spaces, punctuations, (usually) numbers
- Vocabulary Reduction
  - Normalize letter case
    - All converted into lowercase
  - ***Stemming***
    - Remove suffixes: e.g., “announces”, “announced”, “announcing” all to “announc”
    - Plural form to singular form
  - Remove ***stopwords***
    - E.g., a, the, and, or, on…
  - Frequency Filter
    - Remove too rare or too common terms (by setting lower/upper limits)

Preprocessing Text (Cont'd)

```
> stopwords("english")
  [1] "i"          "me"         "my"         "myself"     "we"         "our"
  [7] "ours"       "ourselves"  "you"        "your"       "yours"      "yourself"
 [13] "yourselves" "he"         "him"        "his"        "himself"    "she"
 [19] "her"        "hers"       "herself"    "it"         "its"        "itself"
 [25] "they"       "them"       "their"      "theirs"     "themselves" "what"
 [31] "which"      "who"        "whom"       "this"       "that"       "these"
 [37] "those"      "am"         "is"         "are"        "was"        "were"
 [43] "be"         "been"       "being"      "have"       "has"        "had"
 [49] "having"     "do"         "does"       "did"        "doing"      "would"
 [55] "should"     "could"      "ought"      "i'm"        "you're"     "he's"
 [61] "she's"      "it's"       "we're"      "they're"    "i've"       "you've"
 [67] "we've"      "they've"    "i'd"        "you'd"      "he'd"       "she'd"
 [73] "we'd"       "they'd"     "i'll"       "you'll"     "he'll"      "she'll"
 [79] "we'll"      "they'll"    "isn't"      "aren't"     "wasn't"     "weren't"
 [85] "hasn't"     "haven't"    "hadn't"     "doesn't"    "don't"      "didn't"
 [91] "won't"      "wouldn't"   "shan't"     "shouldn't"  "can't"      "cannot"
 [97] "couldn't"   "mustn't"    "let's"      "that's"     "who's"      "what's"
[103] "here's"     "there's"    "when's"     "where's"    "why's"      "how's"
[109] "a"          "an"         "the"        "and"        "but"        "if"
[115] "or"         "because"    "as"         "until"      "while"      "of"
[121] "at"         "by"         "for"        "with"       "about"      "against"
[127] "between"    "into"       "through"    "during"     "before"     "after"
[133] "above"      "below"      "to"         "from"       "up"         "down"
[139] "in"         "out"        "on"         "off"        "over"       "under"
[145] "again"      "further"    "then"       "once"       "here"       "there"
[151] "when"       "where"      "why"        "how"        "all"        "any"
[157] "both"       "each"       "few"        "more"       "most"       "other"
[163] "some"       "such"       "no"         "nor"        "not"        "only"
[169] "own"        "same"       "so"         "than"       "too"        "very"
```

```
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```

	Docs			
Terms	1	2	3	4
first	1	0	0	0
forth	0	0	0	1
second	0	1	0	0
sentenc	1	1	1	1
third	0	0	1	0

Shmueli, G., Bruce, P., Yahav, I., Patel, N. & Lichtendahl, K. (2017). *Data Mining for Business Analytics: Concepts, Techniques, and Applications in R*. John Wiley & Sons. p 485.

TF-IDF (1/3)

- Value representation in Term-Document Matrix
  - **Presence/Absence**: binary (e.g., vendor names in forensic accounting)
  - **Frequency**: repeated appearance may signal importance
  - **TF-IDF:** identify documents with frequent occurrences of rare terms

- Term Frequency (TF)
  - Document d, term t

$$TF(t, d) = \#\text{ times term } t \text{ appears in document } d$$

  - ***Normalize*** with respect to ***document length***
    - Divide by the total number of terms in document d
  - Measure the relevance of a term to a document

TF-IDF (2/3)

- Inverse Document Frequency (IDF)
  - Measure the rarity of a term in the corpus

$$IDF(t) = \log\left(\frac{\text{total number of documents}}{\text{\# document containing term } t}\right)$$

- TF-IDF

$$TF\text{–}IDF(t, d) = TF(d, t) \times IDF(t)$$

  - High value for a document with a high frequency for a term relatively rare overall
  - Low value for terms <u>*almost absent from a document*</u> or <u>*present in most documents*</u>

TF-IDF (3/3)

- Example

$$TF\text{–}IDF(first, 1) = \frac{1}{2} \times \log_2\left(\frac{4}{1}\right) = 1$$

Terms	Docs 1	2	3	4
first	1	0	0	0
forth	0	0	0	1
second	0	1	0	0
sentenc	1	1	1	1
third	0	0	1	0

Terms	Docs 1	2	3	4
first	1	0	0	0
forth	0	0	0	1
second	0	1	0	0
sentenc	0	0	0	0
third	0	0	1	0

From Terms to Concepts

- From *Terms* to *Concepts (Topics)*
  - Dimension reduction: reduce the large number of terms to a small number of "concepts" or "topics"
  - Each concept involves a cluster of terms, a linear combination of terms: concept-term matrix
  - Each document is represented as a vector of weights associated with different concepts
  - Replace the document-term matrix by document-concept matrix
  - Concepts in statistical sense, not necessarily understandable to humans

- Topic Models
  - Matrix factorization methods: Latent Semantic Analysis (LSA)
  - Probabilistic Topic Models: Latent Dirichlet Allocation (LDA)

From Terms to Concepts (Cont'd)

- Apply all machine learning methods we have learned
  - Attach an *outcome variable* to each document
  - Features (or concepts) of documents serve as the *predictor variables*
  - Clustering
  - Classification
  - Regression/prediction

Summary

- Overview of Text Mining
- Bag of words
- Preprocessing text
- TF-IDF
- Represent documents in concepts or topics

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Topic Modeling

Learning Objectives

- Describe the Latent Dirichlet Allocation (LDA) framework
- Estimate LDA models
- Interpret LDA outputs

Probabilistic Topic Modeling

- Probabilistic Topic Modeling
  - Statistical methods that analyze the words of the original texts to discover the *underlying semantic structure* of a document collection
  - Provide a statistical solution to the problem of managing large archives of documents
  - Summarize and understand the growing digitized archive of information at a scale that would be impossible by human annotation
  - Do NOT require any prior annotations or labeling of the documents – the topics emerge from the analysis of the original texts
  - Essentially *unsupervised* learning, can be combined with *supervised* analysis

- Methods and History
  - Latent Semantic Analysis (LSA) (Deerwester et al, 1990)
  - Probabilistic Latent Semantic Analysis (pLSA) (Hofmann 1999)
  - Latent Dirichlet Allocation (LDA) (Blei et al, 2003)

Latent Dirichlet Allocation (LDA)

- Latent Dirichlet Allocation (LDA)
  - Each *document* exhibits multiple *topics*
  - Each *topic* is a *distribution over a vocabulary of terms*
  - Each *word* in a document is *assigned to a topic*

Topics

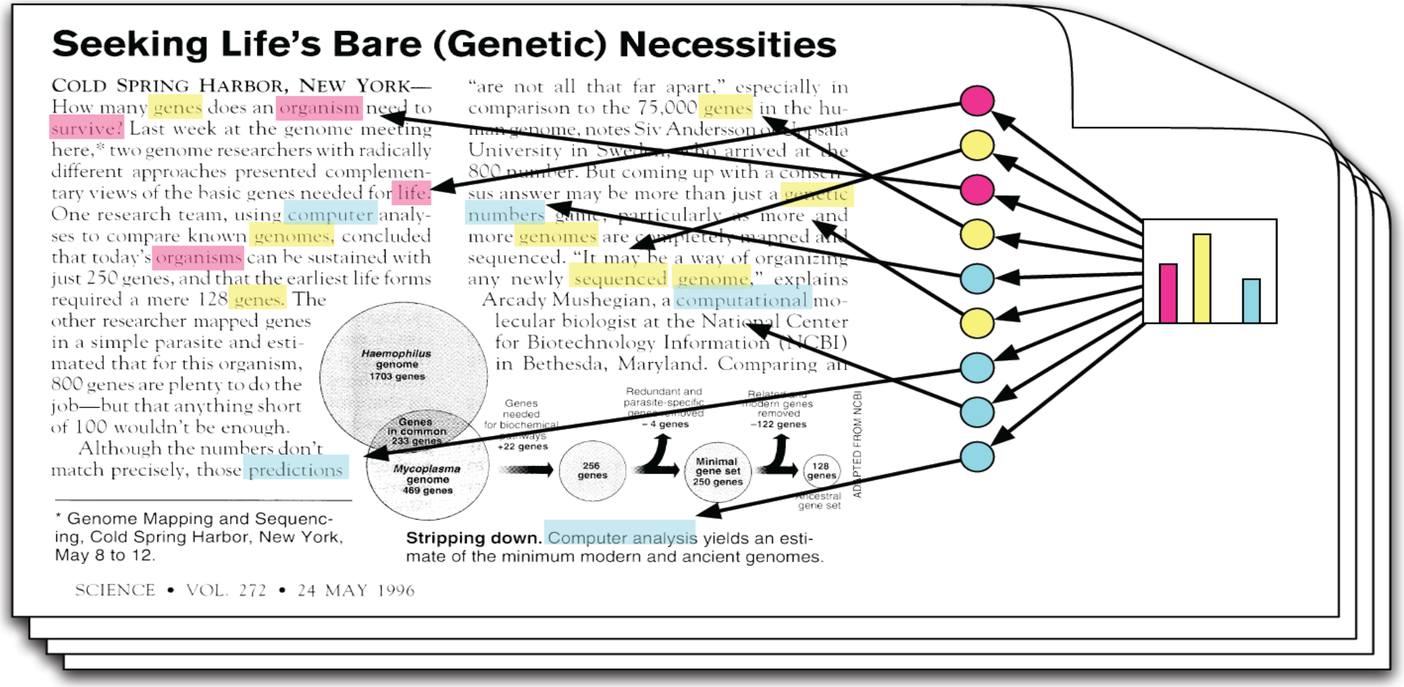
```
gene      0.04
dna       0.02
genetic   0.01
...
```

```
life      0.02
evolve    0.01
organism  0.01
...
```

```
brain     0.04
neuron    0.02
nerve     0.01
...
```

```
data      0.02
number    0.02
computer  0.01
...
```

Documents



Seeking Life's Bare (Genetic) Necessities

COLD SPRING HARBOR, NEW YORK—How many genes does an organism need to survive? Last week at the genome meeting here,* two genome researchers with radically different approaches presented complementary views of the basic genes needed for life. One research team, using computer analyses to compare known genomes, concluded that today's organisms can be sustained with just 250 genes, and that the earliest life forms required a mere 128 genes. The other researcher mapped genes in a simple parasite and estimated that for this organism, 800 genes are plenty to do the job—but that anything short of 100 wouldn't be enough.

Although the numbers don't match precisely, those predictions "are not all that far apart," especially in comparison to the 75,000 genes in the human genome, notes Siv Andersson of Uppsala University in Sweden, who arrived at the 800 number. But coming up with a consensus answer may be more than just a genetic numbers game, particularly as more and more genomes are completely mapped and sequenced. "It may be a way of organizing any newly sequenced genome," explains Arcady Mushegian, a computational molecular biologist at the National Center for Biotechnology Information (NCBI) in Bethesda, Maryland. Comparing an

* Genome Mapping and Sequencing, Cold Spring Harbor, New York, May 8 to 12.

Stripping down. Computer analysis yields an estimate of the minimum modern and ancient genomes.

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Topic proportions and assignments

Latent Dirichlet Allocation (LDA) (Cont'd)

- Latent Dirichlet Allocation (LDA)
  - K topics: each *topic* is represented by β_k: a distribution over a fixed vocabulary V
    - β_k is a V-dimensional vector; all elements add up to 1
  - D documents: the dth *document* is represented by θ_d, where $\theta_{d,k}$ is the *topic proportion* for topic k in document d
    - θ_d is a K-dimensional vector; all elements add up to 1
  - z_d is the *topic assignments* of document d, where $z_{d,n}$ is the topic assignment for the nth *word* in document d
    - z_d is a N-dimensional vector (where N is the total number of words in document d); $z_{d,n} = 1, \dots, K$
  - w_d is the *observed words* of document d, where $w_{d,n}$ is the nth word in document d
  - ***Observed*** data: $w_{1:D}$
  - ***Hidden*** topic structure:
    - The topics $\beta_{1:K}$
    - Per-document topic distributions $\theta_{1:D}$
    - Per-document per-word topic assignments $z_{1:D}$

LDA Generative Process

- Data Generative Process
  - An imaginary random process that is assumed to have produced the observed data

1. For each topic, randomly draw a distribution over words $\beta_k \sim Dir_V(\eta)$
2. For each document, randomly draw a distribution over topics $\theta_d \sim Dir_K(\alpha)$
3. For each word in the document,
   1) Randomly draw a topic from the distribution in 2, $z_{d,n} \sim Mult(\theta_d)$
   2) Randomly draw a word from the distribution in 1, $w_{d,n} \sim Mult(\beta_{z_{d,n}})$

LDA Generative Process (Cont'd)

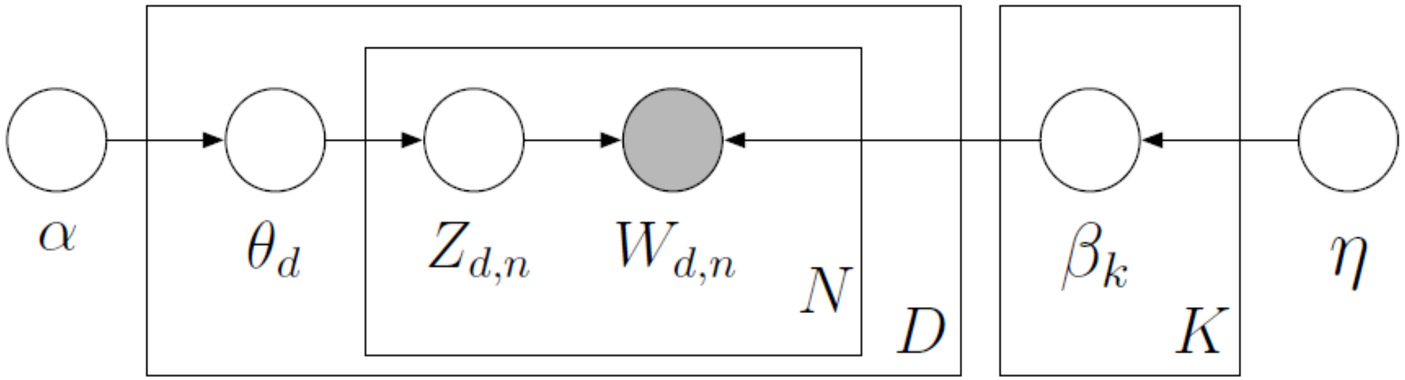
- Dirichlet distribution
  - *A **distribution of** discrete **distributions***
  - Exponential family distribution *over the simplex of positive vectors* that sum to one $(\pi_1, \dots, \pi_K)$

$$p(\pi_1, \dots, \pi_K | \alpha) = \frac{\Gamma(\sum \alpha_i)}{\prod \Gamma(\alpha_i)} \prod_i \pi_i^{\alpha_i - 1}$$

  - $\Gamma(\cdot)$: *Gamma* function, real-valued extension of the *factorial* function
  - Symmetric Dirichlet distribution: $\alpha_1 = \cdots = \alpha_V = \eta$

LDA Graph Model

- Probabilistic Graph Model



- Plate K: each topic
- Plate D: each document
- Plate N: each word
- $w_{d,n}$: actual words, observed data
- $\beta_k, \theta_d, z_{d,n}$: latent topic structure, unobserved and to be inferred
- α, η: parameters of the Dirichlet distributions

Estimating LDA Models (1/3)

- **Bayes' Theorem**
  - Posterior distribution: $p(\beta|D)$
  - Prior distribution: $p(\beta)$
  - Likelihood: $p(D|\beta)$

$$
\begin{aligned}
p(\beta|D) &= \frac{p(\beta, D)}{p(D)} \\
&= \frac{p(D|\beta) \cdot p(\beta)}{\int p(D|\beta) p(\beta)\, d\beta} \\
&\propto p(D|\beta) \cdot p(\beta)
\end{aligned}
$$

  - The *posterior distribution* is proportional to the *likelihood* times the *prior distribution*

Estimating LDA Models (2/3)

- Posterior distribution of LDA parameters

$$
\begin{aligned}
p(\beta_{1:K}, \theta_{1:D}, z_{1:D} | w_{1:D}) &= \frac{p(\beta_{1:K}, \theta_{1:D}, z_{1:D}, w_{1:D})}{p(w_{1:D})} \\
&\propto p(z_{1:D}, w_{1:D} | \beta_{1:K}, \theta_{1:D}) \cdot p(\beta_{1:K}) \cdot p(\theta_{1:D}) \\
&\propto p(w_{1:D} | \beta_{1:K}, z_{1:D}) \cdot p(z_{1:D} | \theta_{1:D}) \cdot p(\beta_{1:K}) \cdot p(\theta_{1:D}) \\
&= \prod_{d=1}^{D} \prod_{n=1}^{N_d} p(w_{d,n} | \beta_{1:K}, z_{d,n}) p(z_{d,n} | \theta_d) \prod_{i=1}^{K} p(\beta_i) \prod_{d=1}^{D} p(\theta_d)
\end{aligned}
$$

Estimating LDA Models (3/3)

- Approximate the posterior distribution
  - Sampling based methods:
    - *Gibbs* or *Metropolis* sampling
    - *Markov Chain Monte Carlo (MCMC)* methods
    - Sequentially sample values following a Markov chain
    - The limiting distribution converges to the target posterior distribution
    - The sample drawn following the Markov chain is *as if* it's drawn from the target posterior distribution
    - The sampled values thus give us an approximation to the *entire distribution* of the unknown parameters, rather than just *point estimates*

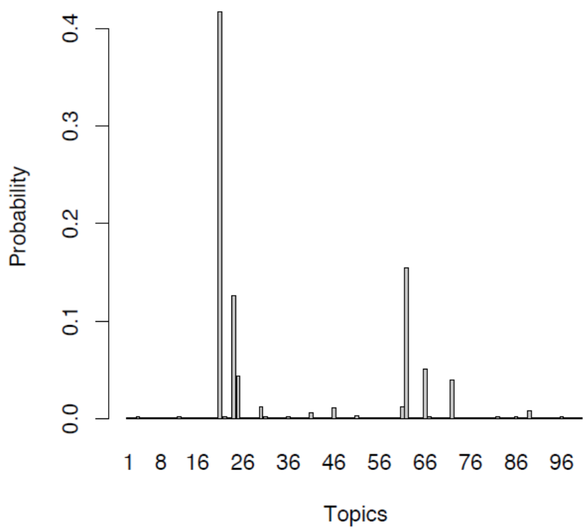
LDA Outputs

- LDA outputs
  - $\boldsymbol{\theta}$ matrix: how each ***document*** is distributed over all ***topics***

    D rows $\times$ K columns
  - $\boldsymbol{\beta}$ matrix: how each ***topic*** is distributed over all ***terms***

    K rows $\times$ V columns

Topic distribution of a document



Top terms related to each topic

"Genetics"	"Evolution"	"Disease"	"Computers"
human	evolution	disease	computer
genome	evolutionary	host	models
dna	species	bacteria	information
genetic	organisms	diseases	data
genes	life	resistance	computers
sequence	origin	bacterial	system
gene	biology	new	network
molecular	groups	strains	systems
sequencing	phylogenetic	control	model
map	living	infectious	parallel
information	diversity	malaria	methods
genetics	group	parasite	networks
mapping	new	parasites	software
project	two	united	new
sequences	common	tuberculosis	simulations

Applications of LDA

- Wide applications of topic modeling to text data
  - Information retrieval, document indexing and search
  - Content recommendation
  - Understand customer feedback and requests
  - Categorization and archival of historical documents, academic literatures, etc.

Applications of LDA (Cont'd)

- Applying LDA to other data
  - Population genetics
    - Genetic patterns within the population: "topics"
    - Each individual expresses different patterns: "documents"
  - Computer vision and image processing
    - Preprocess images to form a collection of "visual words": "terms"
    - "Visual patterns" occurring across a collection of images: "topics"
    - Each image exhibits a combination of visual patterns: "documents"
    - Classify images, label images, search images

Summary

- LDA framework
- Estimating LDA models
- Interpreting LDA outputs
- LDA applications